

Quadratic Functions

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1. [March US 2023]

$$4x^2 + kx + 9 = 0$$

In the given equation, k is a positive constant. The equation has exactly one real solution. What is the value of k ?

2. [March US 2023]

$$h(t) = -4.9t^2 + 10t$$

The function h models the height $h(t)$, in meters, of a football t seconds after it is kicked. What is the interpretation of $h(2) = 0.40$ in this context?

- A) The football has a maximum height of 0.40 meter
- B) The football has a maximum height of 2 meters
- C) The football has a height of 0.40 meter 2 seconds after it is kicked
- D) The football has a height of 2 meters 0.40 second after it is kicked

3. [March US 2023]

$$x^2 - x - 12 = 0$$

What is the sum of the solutions to the given equation?

- A) -7
- B) -1
- C) 1
- D) 7

4. [March US 2023]

$$3x^2 - 2x - 7 = 0$$

What is the sum of the solutions to the given equation?

5. [June 2023]

$$18x^2 + 24x + c = 0$$

In the given equation, c is a constant. The equation has exactly one solution. What is the value of c ?

6. [June 2023]

The function g is a quadratic function. In the xy -plane, the graph of $y = g(x)$ has a vertex at $(-1, -4)$ and passes through the points $(-2, -43)$ and $(1, -160)$. What is the values of $g(0) - g(2)$?

- A) -121
- B) 0
- C) 117
- D) 312

7. [June 2023]

An auditorium has seats for 3,200 people. Tickets to attend a show at the auditorium currently cost \$8.00. For each \$1.00 increase to the ticket price, 100 fewer tickets will be sold. This situation can be modeled by the equation $-100x^2 + 2,400x = 25600$ where x represents the increase in ticket price, in dollar, and y represents the revenue, in dollars, from ticket sales. If the equation is graphed in the xy -plane, at what value of x is the maximum of the graph?

- A) 32 C) 24
B) 12 D) 8

8. [June 2023]

$$\frac{x+1}{5x^2} = \frac{k}{x}$$

In the given equation, k is a constant. The solution to the given equation is $\frac{1}{174}$. What is the value of k ?

9. [June 2023]

If $(x+3)^2 = 30$, what is the value of $x^2 + 6x$?

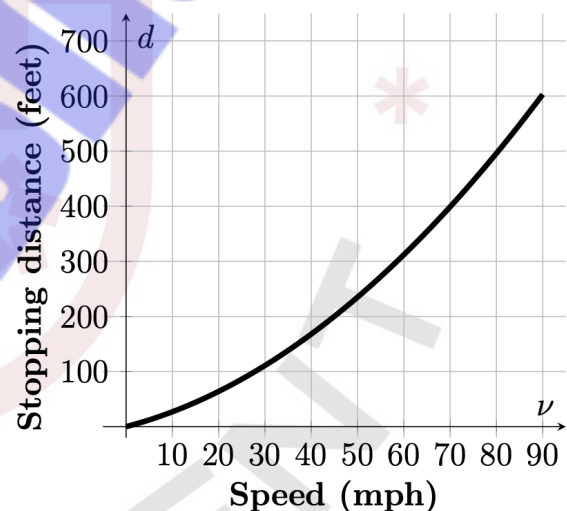
- A) 21 C) 39
B) 30 D) 60

10. [August US 2023]

What is the solution to the equation $2\sqrt{x} = 3$?

- A) 1 C) $\frac{9}{4}$
B) $\frac{3}{2}$ D) $\frac{9}{2}$

11. [August US 2023]



The graph quadratic function shown relates a car's stopping distance d , in feet, to its initial speed v , in miles per hour (mph). Which of the following could be an equation for d ?

- A) $d = 2.2v + 0.05v^2$
B) $d = 2.2v - 0.05v^2$
C) $d = 100 + 2.2v + 0.05v^2$
D) $d = 100 + 2.2v - 0.05v^2$

12. [August US 2023]

$$x^2 - 4x - 1 = 0$$

A solution to the given equation is $x = \sqrt{k} + 2$. What is the value of k ?

13. [August US 2023]

$$3x^2 - 4x + 7 = 0$$

How many distinct real solutions does the given equation have

- A) Zero
- B) Exactly one
- C) Exactly two
- D) Infinitely many

15. [November 2023]

The solutions to $x^2 + 10x + 23 = 0$ are r and s , where $r < s$. The solutions to $x^2 + 10x + 17 = 0$ are t and u , where $t < u$. The solutions to $x^2 + 20x + c = 0$, where c is a constant are $r + u$ and $s + t$. What is the value of c ?

16. [December 2023]

$$\frac{1}{x} + \frac{1}{x-3} = \frac{6}{x^2 - 3x}$$

What are all the solutions to the given equation?

- | | |
|-------------|------------------|
| A) -1 and 6 | C) 3 |
| B) 0 and 3 | D) $\frac{9}{2}$ |

14. [November 2023]

In the equation $10x^2 + 50x + c = 0$, c is a constant. If the equation has exactly one solution, what is the value of c ?

- | | |
|------------------|--------------------|
| A) 0 | C) 60 |
| B) $\frac{5}{2}$ | D) $\frac{125}{2}$ |

17. [December 2023]

$$x^2 - 12x + c = 0$$

In the given equation, c is a constant. The equation has no real solutions if $c > n$. What is the least possible value of n ?

- | | |
|--------|-------|
| A) -36 | C) 12 |
| B) -12 | D) 36 |

18. [December 2023]

$$22x^2 + (22s + r)x + rs = 0$$

In the given equation, r and s are positive constants. The product of the solutions to the given equation is krs , where k is a constant. What is the value of k ?

19. [December 2023]

$$y = -4x^2 - 23$$

$$y = qx - 19$$

In the given system of equations, q is a positive constant. The graphs of the equations in the given system intersect at exactly one point (x, y) , in the xy -plane. What is the value of x ?

A) -42

B) -27

C) -8

D) -1

20. [March 2024]

What value of x is the solution to the equation $65x + 20 = -15x^2$?

A) $\frac{1}{4}$

B) $\frac{1}{3}$

C) 3

D) 5

21. [March 2024]

$$\sqrt{k - x} = 52 - x$$

In the given equation, k is a constant. The equation has exactly one real solution. What is the minimum possible value of $4k$?

22. [March 2024]

The function f is defined by

$$f(x) = ax^2 + bx + c$$

where a, b , and c are constants and $1 < a < 4$. The graph of $y = f(x)$ in the xy -plane passes through the points $(12, 0)$ and $(-4, 0)$. If a is an integer, what could be the value of $a + b$?

23. [March 2024]

$$x^2 - 10x + 24 = 0$$

How many distinct real solutions does the given equation have?

A) Exactly one

B) Exactly two

C) Infinitely many

D) Zero

24. [March 2024]

$$y = -\frac{1}{4}x^2 + x + 28$$

The given equation models a company's active projects over 6 months, where y is the estimated number of active projects x months after the end of April 2012, where $0 \leq x \leq 6$. Which statement is the best interpretation of the y -intercept of the graph of this equation in the xy -plane?

- A) At the end of April 2012, the estimated number of active projects was 0.
- B) At the end of April 2012, the estimated number of active projects was 28.
- C) At the end of May 2012, the estimated number of active projects was 0.
- D) At the end of May 2012, the estimated number of active projects was 28.

25. [March 2024]

$$-16(5x - 3)^2 + 4(5x - 4)^2$$

The given expression can be rewritten as

$$\frac{a}{6}x^2 + \frac{b}{6}x + \frac{c}{6},$$

where a , b , and c are constants. What is the value of $a + b + c$?

- A) -360
- B) -180
- C) -60
- D) -10

26. [March 2024]

Function f is a quadratic function. The graph of $y = f(x)$ in the xy -plane has a vertex at $(-8, 1)$, contains the point $(9, -1)$, and has a y -intercept at $(0, a)$. The graph of $y = 6 \cdot f(x)$ has a y -intercept at $(0, b)$. What is the positive difference between a and b ?

27. [March 2024]

$$-16(5x - 2)^2 + 6(5x - 3)^2$$

The given expression can be rewritten as

$$\frac{a}{6}x^2 + \frac{b}{6}x + \frac{c}{6},$$

where a , b , and c are constants. What is the value of $a + b + c$?

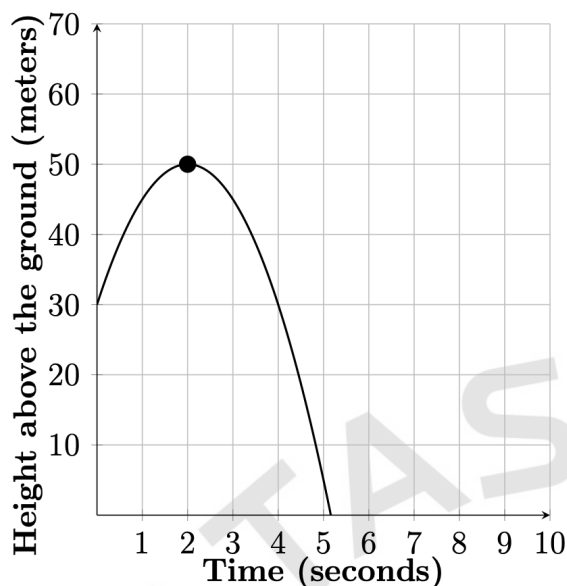
- A) -720
- B) -360
- C) -120
- D) -20

28. [March 2024]

$$\sqrt{k - x} = 54 - x$$

In the given equation, k is a constant. The equation has exactly one real solution. What is the minimum possible value of $4k$?

29. [March 2024]



An object was launched upward from a platform. The graphs shown models the height, y , of the object x seconds after it was launched. For which of the following intervals of time was the height of the object increasing for the entire interval?

- A) From $x=0$ to $x=2$
- B) From $x=0$ to $x=4$
- C) From $x=2$ to $x=3$
- D) From $x=4$ to $x=5$

30. [March 2024]

$$f(x) = 14x^2 - 35x + 7$$

The given equation defines the function f . For what values of x does $f(x)$ reach its minimum?

- A) 1.25
- B) 1.75
- C) 7
- D) 35

31. [March 2024]

An object is kicked from a platform. The equation $h = -4.9t^2 + 13t + 15$ represents this situation, where the h is the height of the object above the ground, in meters, t seconds after it is kicked. Which number represents the height, in meters, from which the object was kicked?

- A) 0
- B) 4.9
- C) 13
- D) 15

32. [March 2024]

The function f is defined by $f(x) = ax^2 + bx + c$, where a , b , and c are constants and $1 < a < 4$. The graph of $y = f(x)$ in the xy -plane passes through the points $(11, 0)$, and $(-5, 0)$. If a is an integer, what could be the value of $a + b$?

33. [May 2024]

The y -intercept of the graph of $y = 2x^2 + 27$ is $(0, y)$. What is the value of y ?

34. [March 2024]

$$y = 6x^2 - 48x + 98$$

$$y + 4 = 0$$

How many solutions are there to the given system of equations have?

- A) There is exactly 1 solution.
- B) There are exactly 2 solutions.
- C) There are exactly 3 solutions.
- D) There are no solutions.

37. [May US 2024]

$$y = x - c$$

$$y = -8(x - 10)^2$$

In the given system of equations, c is a constant. The system has two distinct real solutions. Which of the following could be the value of c ?

- A) 5
- B) 9
- C) $\frac{319}{32}$
- D) 15

35. [May 2024]

$$y = 5(x - 3)$$

$$y = (x - 3)^2$$

Which ordered pair (x, y) is a solution to the given system of equations?

- A) $(8, -5)$
- B) $(8, 25)$
- C) $(3, 8)$
- D) $(2, -5)$

38. [May US 2024]

$$y = -\frac{1}{4}x^2 + x + 29$$

The given equation models a company's active projects 6 months, where y is the estimated number of active projects x months after the end of April 2013, where $0 \leq x \leq 6$. Which statement is the best interpretation of the y -intercept of the graph of this equation in the xy -plane?

36. [May US 2024]

The function f is a quadratic function. In the xy -plane, the graph of $y = f(x)$ has a vertex at $(1, 9)$ and passes through the points $(2, 33)$ and $(-1, 105)$. What is the value of $f(-2) - f(0)$?

- A) 81
- B) 129
- C) 192
- D) 225

- A) At the end of April 2013, the estimated number of active projects was 0.
- B) At the end of April 2013, the estimated number of active projects was 29.
- C) At the end of May 2013, the estimated number of active projects was 0.
- D) At the end of May 2013, the estimated number of active projects was 29.

39. [May US 2024]

$$x^2 + 11x + 19 = 0$$

One solution to the given equation can be written as $x = \frac{-11 - \sqrt{k}}{2}$, where k is a constant. What is the value of k ?

40. [May 2024]

$$f(x) = \sqrt{5x + 6}$$

The function f is defined by the given equation. If $f(a) = -5a$, where a is a constant, what is the value of a ?

- A) $\frac{3}{5}$
B) $\frac{2}{5}$

- C) $-\frac{2}{5}$
D) $-\frac{3}{5}$

41. [May 2024]

$$x^2 = 2,601$$

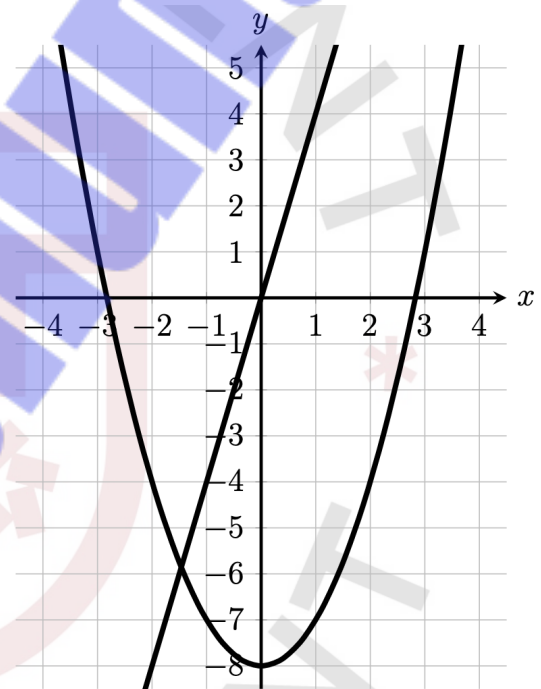
How many distinct real solutions does the given equation have?

- A) Zero
B) Exactly one
C) Exactly two
D) Infinitely many

42. [May 2024]

The expression $(9x - 28)(15x + 4)$ is equivalent to the expression $ax^2 + bx + c$, where a , b , and c are constants. What is the value of b ?

43. [May US 2024]



The graph of a system of a linear and a quadratic equation is shown. Which system of equations is represented by the graph?

- | | |
|-------------------------------|------------------------------|
| A) $y = -4x$
$y = x^2 + 8$ | C) $y = 4x$
$y = x^2 + 8$ |
| B) $y = -4x$
$y = x^2 - 8$ | D) $y = 4x$
$y = x^2 - 8$ |

44. [May 2024]

Which of the following expressions is (are) a factor of $4x^2 + 31x - 45$?

I. $x - 9$

II. $4x - 5$

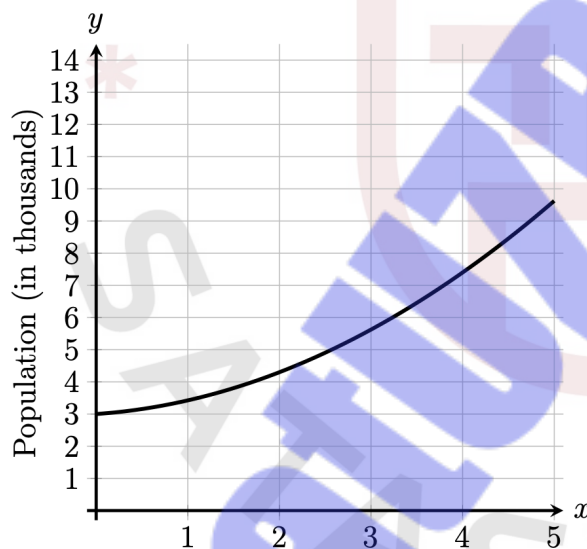
- A) I only
- B) II only
- C) I and II
- D) Neither I nor II

46. [May 2024]

Which quadratic equation has exactly one distinct real solution?

- A) $x^2 - 22 = 0$
- B) $x^2 - 11x = 0$
- C) $x^2 - 11x + 121 = 0$
- D) $x^2 - 22x + 121 = 0$

45. [May 2024]



In 2000, a census was taken to determine the population of a certain town. The graph gives the estimated population y , in thousands, of this town x years after the 2000 census was taken, where $0 \leq x \leq 5$. Based on the graph, which of the following is the closest to the estimated population 2 years after the 2000 census was taken?

- A) 7,400
- B) 6,200
- C) 4,300
- D) 3,000

47. [August US 2024]

The function f is defined by

$$f(x) = -5x^2 - 7.$$

What is the value of $f(-1)$?

- A) -12
- B) -2
- C) 2
- D) 12

48. [May 2024]

$$x(kx - 68) = -4$$

In the given equation, k is an integer constant. If the equation has no real solution, what is the least possible value of k ?

49. [May 2024]

$$f(x) = x + 4$$

$$g(x) = 3x^2 - rx + 48$$

The functions f and g are given. In function g , r is a constant. If $f(x) \cdot g(x) = 3x^3 + 192$, what is the value of r ?

52. [June 2024]

If $4x^2 - 24x - 36 = 0$, what is the value of $x^2 - 6x$?

A) 9

C) 4

B) 6

D) 0

50. [August US 2024]

$$y = -0.5$$

$$y = x^2 + 10x + a$$

In the given system of equations, a is a positive integer constant. The system has no real solutions. What is the least possible value of a ?

53. [June 2024]

The function $f(x) = \frac{1}{9}(x - 4)^2 + 5$ gives a toy car's height above the ground $f(x)$ in inches, x seconds after it started moving on an elevated track, where $0 \leq x \leq 10$. Which of the following is the best interpretation of the vertex of the graph of $y = f(x)$ in the xy -plane?

A) The toy car's minimum height was 5 inches above the ground.

B) The toy car's minimum height was 4 inches above the ground.

C) The toy car's height was 5 inches above the ground when it started moving.

D) The toy car's height was 4 inches above the ground when it started moving.

51. [August US 2024]

$$y = 2x^2 - 19x + 40$$

$$y = x + a$$

In the given system of equations, a is a constant. The graph of the equations in the given system intersect at exactly one point (x, y) , in the xy -plane. What is the value of x ?

A) -10

C) 5

B) -5

D) 10

54. [August US 2024]

The function f is defined by

$$f(x) = ax^2 + bx + c,$$

where a, b and c are constants. The graph of $y = f(x)$ in the xy -plane passes through the points $(9,0)$ and $(-2,0)$. If a is an integer greater than 1, which of the following could be the value of $a + b$?

- A) 8
- B) 7
- C) -6
- D) -12

57. [June 2024]

What is the y -intercept of the graph of $y = 2x^2 + 6x + 3$ in the xy -plane?

- A) $(0, 2)$
- B) $(0, 3)$
- C) $(0, 6)$
- D) $(0, 9)$

55. [August US 2024]

A quadratic function models the height, in feet, of an object above the ground in terms of the time, in seconds, after the object was launched. According to the model, the object was launched from a height of 0 feet and reached its maximum height of 1,600 feet 10 seconds after it was launched. Based on the model, what was the height, in feet, of the object 15 seconds after it was launched?

58. [August 2024]

$$y = x^2 - 18x + 51$$

The given equation relates the variables x and y . For what value of x does the value of y reach its minimum?

- A) -18
- B) 9
- C) 18
- D) 51

56. [June 2024]

$$f(x) = (x - 2)^2 + 9$$

What is the minimum value of the given function?

- A) 2
- B) 7
- C) 9
- D) 11

59. [June 2024]

$$\frac{1}{cx} = \frac{x}{152} + \frac{1}{c}$$

In the given equation, c is a constant. If the equation has exactly one solution, what is the value of c ?

60. [June 2024]

$$y = 2x^2 - 23x + 64$$

$$y = x + a$$

In the given system of equations, a is a constant. The graphs of the equations in the given system intersect at exactly one point, (x, y) , in the xy -plane. What is the value of x ?

- A) -8
B) -6
C) 6
D) 8

61. [June 2024]

If $4x^2 - 12x - 24 = 0$, what is the value of $x^2 - 3x$?

- A) 0
B) 3
C) 4
D) 6

62. [June 2024]

$$f(x) = (x - 4)^2 + 11$$

What is the minimum value of the given function?

- A) 4
B) 7
C) 11
D) 15

63. [June 2024]

For the quadratic function f , the table shows three values of x and their corresponding values of $f(x)$. Which equation defines f ?

x	$f(x)$
-1	4
0	12
1	22

- A) $f(x) = 5x^2 + 5x + 12$
B) $f(x) = 9x^2 + x + 12$
C) $f(x) = 9x^2 - x + 12$
D) $f(x) = x^2 + 9x + 12$

64. [June 2024]

The function f is defined by $f(x) = 4x^2$. What is the value of $f(11)$?

- A) 44
B) 52
C) 88
D) 484

65. [June 2024]

$$-4x^2 + bx - 25 = 0$$

In the given equation, b is a positive integer. The equation has no real solution. What is the largest possible value of b ?

66. [August 2024]

The function f is defined by $f(x) = 5x^2$. What is the value of $f(6)$?

- A) 30 C) 60
B) 40 D) 180

67. [August 2024]

An object is kicked from a platform. The equation $h = -4.9t^2 + 5t + 9$ represents this situation, where h is the height of the object above the ground, in meters, t seconds after it is kicked. Which number represents the height, in meters, from which the object was kicked?

- A) 0 C) 5
B) 4.9 D) 9

68. [August 2024]

$$3x^2 + x - 5 = 0$$

What is the greatest solution to the given equation?

- A) $-\frac{1}{6} - \frac{\sqrt{61}}{6}$
B) $-\frac{1}{6} + \frac{\sqrt{61}}{6}$
C) $-\frac{1}{3} - \frac{\sqrt{61}}{3}$
D) $-\frac{1}{3} + \frac{\sqrt{61}}{3}$

69. [August 2024]

$$33x^2 + (33s + r)x + rs = 0$$

In the given equation, r and s are positive constants. The product of the solutions to the given equation is krs , where k is a constant. What is the value of k ?

70. [August 2024]

$$f(x) = -8x^2 + 101x$$

The function f models the depth below sea level, in meters, of a certain seal x minutes after the seal dives into the water. What is the best interpretation of $f(2) = 170$ in this context?

- A) 2 minutes after the seal dives into the water, its estimated depth is 170 meters below the sea level.
B) Approximately 2 minutes after the diving into the water, the sea level returns to sea level.
C) The sea level reaches an estimated maximum depth of 170 meters below sea level.
D) 170 minutes after the seal dives into the water, its estimated depth is 2 meters below the sea level.

71. [August US 2024]

$$(v - 12)(v - 3) = 0$$

What are all possible solutions to the given equation?

- A) -12 and 3
- B) -12 and 9
- C) -3 and 12
- D) 3 and 12

72. [August US 2024]

Which expression is a factor of

$$7x^2 - 9x + 14x - 18?$$

- A) $x - 2$
- B) $x + 9$
- C) $7x - 9$
- D) $7x + 2$

73. [August US 2024]

$$x^2 + y + 3 = 3$$

$$8x + 16 - y = 0$$

The solution to the given system of equations is (x, y) . What is the value of x ?

- A) -16
- B) -4
- C) 2
- D) 8

74. [August 2024]

A machine launches a baseball from ground level. The baseball reaches a maximum height of 92.16 meters above the ground at 2.4 seconds and hits the ground at 4.8 seconds. Which equation represents the height above ground h , in meters, of the baseball t seconds after it is launched?

- (A) $h = -t^2 + 4.8$
- (B) $h = -t^2 + 92.16$
- (C) $h = -16(t - 2.4)^2 - 4.8$
- (D) $h = -16(t - 2.4)^2 + 92.16$

75. [August US 2024]

The function g is defined by

$$g(x) = (x + 17)(t - x),$$

where t is a constant. In the xy -plane, the graph of $y = g(x)$ passes through the point $(23, 0)$. What is the value of $g(0)$?

76. [August US 2024]

$$x^2 + 5x = 47$$

The solution to the given equation are $\frac{m + \sqrt{q}}{2}$ and $\frac{m - \sqrt{q}}{2}$, where m and q are integers. What is the value of $m + q$?

77. [August 2024]

If $(x+3)^2 = 33$, what is the value of $x^2 + 6x$?

78. [August US 2024]

In the equation $7x^2 + 49x + c = 0$, c is a constant. If the equation has exactly one real solution, what is the value of c ?

- A) 0
- B) $\frac{7}{2}$
- C) 56
- D) $\frac{343}{4}$

79. [August US 2024]

$$\sqrt{k-x} = 57-x$$

In the given equation, k is a constant. The equation has exactly one real solution. What is the minimum possible value of $4k$?

80. [August US 2024]

The y -intercept of the graph of $y = x^2 + 38$ in the xy -plane is $(0, y)$. What is the value of y .

81. [August US 2024]

The equation $-4.9(x - 9.1)^2 + 10700$ give the estimated height above the ground, y , in meters, of a plane, where x is the number of second since it started a parabolic maneuver. If this equation is graphed in the xy -plane, which of the following is the best interpretation of the vertex of the graph?

- A) The plane reached an estimated maximum height of 10700 meters 4.9 seconds after it started the parabolic maneuver.
- B) The plane reached an estimated maximum height of 10700 meters 9.1 seconds after it started the parabolic maneuver.
- C) The plane reached an estimated maximum height of 4.9 meters 10700 seconds after it started the parabolic maneuver.
- D) The plane reached an estimated maximum height of 9.1 meters 10700 seconds after it started the parabolic maneuver.

82. [August US 2024]

$$p(x) = 2x^2 + 20x + 15$$

What is the minimum value of the given function?

- | | |
|--------|-------|
| A) -35 | C) -5 |
| B) -15 | D) 15 |

83. [August US 2024]

$$x + 47 = y$$

$$(x + 47)^2 = y$$

A solution to the given system of equations is (x, y) . What is a possible value of x ?

- A) 47
- B) 1
- C) 0
- D) -47

84. [August US 2024]

The function g is defined by

$$g(x) = (x + 17)(t - x),$$

where t is a constant. In the xy -plane, the graph of $y = g(x)$ passes through the point $(23, 0)$. What is the value of $g(0)$?

85. [August US 2024]

$$(x + 81)^2 = 256$$

How many distinct real solutions does the given equation have?

- A) Exactly one
- B) Exactly two
- C) Infinitely many
- D) Zero

86. [October US 2024]

$$(2x - 7)^2 - 8(2x - 7) + 15 = 0$$

What is a solution to the given equation?

87. [October US 2024]

The function $f(x)$ is defined by

$$f(x) = 10x^2 - 38x - 150.$$

What is the value of $f(0)$?

- A) -150
- B) -38
- C) 0
- D) 10

88. [October US 2024]

An object is launched into the air from ground level. According to a quadratic model, 2.1 seconds after the object is launched, it reaches its maximum height of 70.56 feet above ground level. Which equation represents this model, where $f(t)$ is the object's height, in feet, above ground level t seconds after it was launched?

- A) $f(t) = -16(t - 2.1)^2 + 70.56$
- B) $f(t) = 16(t + 2.1)^2 + 70.56$
- C) $f(t) = -16(t + 2.1) + 70.56$
- D) $f(t) = (t - 2.1)^2 + 70.56$

89. [October US 2024]

$$f(x) = 5x^2 + 60x + 181$$

The function g is defined by $g(x) = f(x+8)$. What is the minimum value of $g(x)$?

- A) -14 C) 1
B) -6 D) 9

90. [October US 2024]

$$-2x^2 + bx - 32 = 0$$

In the given equation, b is a positive integer. The equation has no real solution. What is the largest possible value of b ?

91. [October 2024]

$$r(t) = 55t - 2t^2$$

The function r is defined by the given equation. The function s is defined by $s(t) = r(t) + 1$. Which expression represents the maximum value of $s(t)$?

- A) $1 + \left(\frac{55}{2}\right)^2$ C) $1 - 2\left(\frac{55}{4}\right)^2$
B) $1 + 2\left(\frac{55}{4}\right)^2$ D) $1 - \left(\frac{55}{2}\right)^2$

92. [October US 2024]

$$\frac{1}{cx} = \frac{x}{76} + \frac{1}{c}$$

In the given equation, c is a constant. If the equation has exactly one solution, what is the value of c ?

93. [October 2024]

$$y = 0.25x^2 - 6.5x$$

The equation gives the estimated stock price y , in dollars, for a certain company x days after the company went public, where $0 \leq x \leq 20$. Which statement is the best interpretation of $(x, y) = (1, 58)$ in this context?

- A) 1 day after the company went public, the company's estimated stock price is \$58.
B) 58 days after the company went public, the company's estimated stock price is \$1.
C) The company's estimated stock price increased \$58 every day after the company went public.
D) The company's estimated stock price increased \$1 every 58 days after the company went public.

94. [October 2024]

$$x^2 - 5x - 24 = 0$$

What is the sum of the solutions to the given equation?

95. [October US 2024]

$$f(x) = -(x - 6)^2 + 4$$

The function f is defined by the given equation. For what value of x does $f(x)$ reach its maximum?

96. [October 2024]

Function f is a quadratic function where $f(-20) = 0$ and $f(-4) = 0$. The graph of $y = f(x)$ in the xy -plane has a vertex at $(r, -64)$. What is the value of r ?

97. [October 2024]

The function f gives the product of a number, x , and a number that is 91 more than x . Which equation defines f ?

- A) $f(x) = -x^2 + x + 91$
- B) $f(x) = x^2 + 91$
- C) $f(x) = x^2 + 91x$
- D) $f(x) = x^2 + 91x + 91$

98. [October 2024]

$$y = 2x^2 - 15x + 23$$

$$y = x + a$$

In the given system of equations, a is a constant. The graphs of the equations in the given system intersect at exactly one point, (x, y) , in the xy -plane. What is the value of x ?

- A) -9
- B) -4
- C) 4
- D) 9

99. [October US 2024]

$$x(x - 15) = 34$$

What is the positive solution to the given equation?

100. [October US 2024]

$$x^2 - 100 = 0$$

How many distinct real solutions does this given equation have?

- A) Zero
- B) Exactly one
- C) Exactly two
- D) Infinitely many

101. [October US 2024]

$$\frac{x}{(x-1)(x+1)} = \frac{81}{56x}$$

Which of the following is a solution to the given equation?

- A) $\frac{1}{5}$ C) 1
B) $\frac{9}{25}$ D) $\frac{9}{5}$

102. [October US 2024]

$$x(9x - 2a) + b(9x - 2a) - (x + b) = 0$$

In the given equation, a and b are positive constants. A solution to the given equation is $x = 42$. What is the value of a ?

103. [October US 2024]

$$3w^2 + 11w - c = 0$$

In the given equation, c is a constant. A solution to the given equation is $\frac{-11+\sqrt{433}}{6}$. What is the value of c ?

104. [October US 2024]

The function f is defined by

$$f(x) = 3\left(\frac{1}{4} - x\right)^2 + \frac{13}{4}.$$

What is the value of $f\left(\frac{1}{4}\right)$?

105. [October US 2024]

$$f(x) = -16x^2 + 99$$

The function f gives the estimated height, in feet, of an acorn x seconds after the acorn fell from a tree. Based on the function, what is the estimated height, in feet, of an acorn before it fell from the tree?

106. [October US 2024]

The variable x and y have quadratic relationship. When $x = 0$, $y = 45$. When $x = 6$, $y = 81$. When $x = 15$, $y = 0$. Which equation represents this relationship?

- A) $y = x^2 - 18x + 45$
B) $y = -x^2 + 12x + 45$
C) $y = x^2 + 45$
D) $y = -3x^2 - 48x + 45$

107. [October US 2024]

$$\frac{1}{x} + \frac{1}{x-3} = \frac{6}{x^2 - 3x}$$

What are all the solution to the given equation?

- A) -1 and 6 C) 3
B) 0 and 3 D) $\frac{9}{2}$

108. [October US 2024]

$$x^2 - 2x - 18 = 0$$

One solution to the given equation can be written as $1 + \sqrt{k}$, where k is a constant. What is the value of k ?

- A) 17 C) 38
B) 19 D) 76

109. [October US 2024]

For a certain type of rope, the equation $y = 900ax^2$, where a is a constant, gives the estimated breaking strength y , in pounds, of a rope with a circumference of x inches. Based on this equation, if a rope of this type has a circumference of 2.75 inches, it has an estimated breaking strength of 9,528.75 pounds. What is the estimated breaking strength, in pounds, of a rope of this type that has a circumference of 7.50 inches?

110. [October US 2024]

$$4x^2 + 21 = 48 + r$$

In the given equation, r is a constant. The equation has exactly one real solution. What is the value of r ?

111. [October US 2024]

The function f is defined by

$$f(x) = ax^2 + bx + c,$$

where a , b , and c are constants. The graph of $y = f(x)$ in the xy -plane passes through the points $(10, 0)$ and $(-5, 0)$. If a is an integer greater than 1, which of the following could be the value of $a + b$?

- A) -8 A) 5
A) -4 A) 6

112. [November 2024]

$$f(x) = (2x - 20)(2x + 8)$$

What is the minimum value of the given function?

113. [November 2024]

Which expression is equivalent to

$$(w + 7)(3w - 10)?$$

- A) $4w^2 - 3w - 3$
B) $4w^2 + 10w - 3$
C) $3w^2 + 11w - 70$
D) $3w^2 + 21w - 70$

114. [November 2024]

If

$$\left(\frac{1}{2}x - \frac{33}{2}\right)^2 - 64 = 0,$$

what is the value of $(x - 33)^2$?

- A) 8
- B) 33
- C) 128
- D) 256

115. [November 2024]

Which expression is equivalent to

$$(x + 8)^2 + 14(x + 8) + 49?$$

- A) $(x + 15)^2$
- B) $(x + 7)^2$
- C) $8(x + 7)^2$
- D) $7(x + 8)^2$

116. [November 2024]

$$2x^2 - 12x - 13 = 0$$

One solution to the given equation can be written as $h - \frac{1}{2}\sqrt{k}$, where h and k are integer constants. What is the value of $h + k$?

- A) 13
- B) 14
- C) 31
- D) 65

117. [November 2024]

The function $f(x) = \frac{1}{9}(x - 6)^2 + 3$ gives a toy car's height above the ground $f(x)$, in inches, x seconds after it started moving on an elevated track, where $0 \leq x \leq 10$. Which of the following is the best interpretation of the vertex of the graph of $y = f(x)$ in the xy plane?

- A) The toy car's minimum height was 3 inches above the ground.
- B) The toy car's minimum height was 6 inches above the ground.
- C) The toy car's height was 3 inches above the ground when it started moving.
- D) The toy car's height was 6 inches above the ground when it started moving.

118. [November 2024]

$$\frac{1}{cx} = \frac{x}{148} + \frac{1}{c}$$

In the given equation, c is a constant. If the equation has only one solution, what is the value of c ?

119. [November US 2024]

$$x^2 = (40)(40)$$

What is the positive solution to the given equation?

120. [November US 2024]

$$y = 4x + 12$$

$$y = (x + 2)^2$$

A solution to the given system of equation is (x, y) . What is the value of x^2 ?

121. [November US 2024]

$$x(kx - 40) = -8$$

In the given equation, k is an integer constant. If the equation has two distinct real solutions, what is the greatest possible value of k ?

122. [November US 2024]

$$f(x) = -9x^2 + 113x$$

The function f models the depth below sea level, in meters, of a certain seal x minutes after the seal dives into the water. What is the best interpretation of $f(2) = 190$ in this context?

- A) 190 minutes after the seal dives into the water, its estimated depth is 2 meters below sea level.
- B) The seal reaches an estimated maximum depth of 190 meters below sea level.
- C) 2 minutes after the seal dives into the water, its estimated depth is 190 meters below the sea level
- D) Approximately 2 minutes after diving into the water, the seal returns to sea level.

123. [November US 2024]

Which expression is equivalent to

$$(x^2 + 11)^2 + (x - 8)(x + 8)$$

- A) $x^4 + x^2 + 185$
- B) $x^4 + 12x^2 + 121$
- C) $x^4 + 23x^2 - 53$
- D) $x^4 + 23x^2 + 57$

124. [November US 2024]

$$x^2 + 2x = 72$$

What is one of the solutions to the given equation?

- A) $1 + \sqrt{73}$
- B) $1 + \sqrt{71}$
- C) $-1 + \sqrt{73}$
- D) $-1 + \sqrt{71}$

125. [November US 2024]

$$\sqrt{k - x} = 32 - x$$

In the given equation, k is a constant. The equation has exactly one real solution. What is the minimum possible value of $4k$?

126. [November US 2024]

Which of the following expressions has a factor of $x + 2b$, where b is a positive integer constant?

- A) $3x^2 + 7x + 14b$
- B) $3x^2 + 16x + 14b$
- C) $3x^2 + 18x + 14b$
- D) $3x^2 + 25x + 14b$

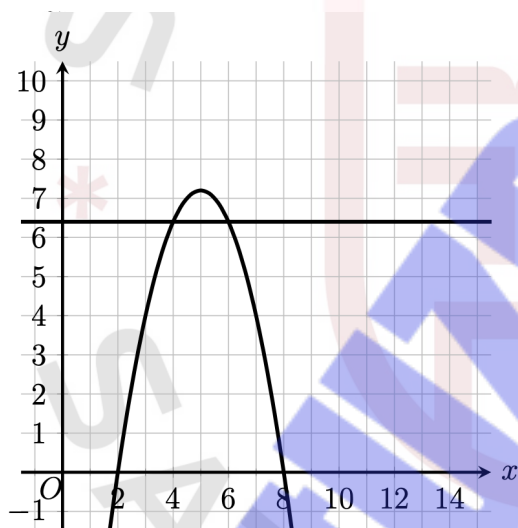
128. [December US 2024]

Which expression is equivalent to

$$13(x^2 - 7)?$$

- A) $13x^2 - 91$
- B) $13x^2 - 20$
- C) $13x^2 - 7$
- D) $13x^2 + 6$

127. [November US 2024]



The graph of a system of a linear equation and a quadratic equation is shown. A solution to the system is (x, y) . What is a possible value of x ?

- A) 4
- B) 5
- C) 6.4
- C) 7

129. [December US 2024]

$$-(3x - 7)^2 = p + 16$$

In the equation, p is an integer constant. The equation has no real solution. What is the least possible value of p ?

- A) -18
- B) -17
- C) -16
- D) -15

130. [December US 2024]

Which quadratic equation has no real solution?

- A) $-2x^2 = 0$
- B) $-2x^2 + 2 = 0$
- C) $-2x^2 + 2x = 0$
- D) $-2x^2 + 2x - 2 = 0$

131. [December 2024]

$$21x^2 + (21s + r)x + rs = 0$$

In the given equation, r and s are positive constants. The product of the solutions to the given equation is krs , where k is a constant. What is the value of k ?

132. [December 2024]

$$-16(5x - 3)^2 + 4(5x - 2)^2$$

The given expression can be rewritten as $\frac{a}{4}x^2 + \frac{b}{4}x + \frac{c}{4}$, where a , b , and c are constants. What is the value of $a + b + c$?

A) -112

B) -56

C) -28

D) -7

134. [December US 2024]

If $121x^2 = -110x + 56$, what is the value of $11x + 5$, where $11x + 5 > 0$?

135. [December 2024]

The function f is defined by $f(x) = 4x^2$. What is the value of $f(11)$?

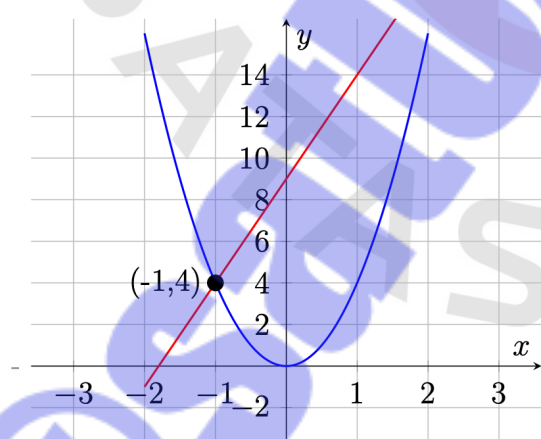
A) 44

B) 52

C) 88

D) 484

133. [December 2024]



The graph of a system of a linear equation and a quadratic equation is shown. Which of the following is a solution (x, y) to the system?

A) $(-1, 4)$

B) $(0, 0)$

C) $(0, 9)$

D)

136. [December 2024]

Which expression is equivalent to

$$2\left(x - \frac{3}{2}\right)(x + 14)?$$

A) $x^2 + 25x - 21$

B) $2x^2 + 11x - 21$

C) $2x^2 + 25x - 42$

D) $2x^2 - 42$

137. [December 2024]

$$-8x(x + 9) = 40$$

One solution to the given equation can be written as $x = -\frac{s+\sqrt{t}}{2}$, where s and t are positive integers. What is the value of $\frac{s}{t}$?

- A) $\frac{9}{101}$
B) $\frac{9}{76}$

- C) $\frac{9}{61}$
D) $\frac{18}{61}$

138. [December 2024]

A quadratic function models the height, in feet, of an object above the ground in terms of the time, in seconds, after the object was launched. According to the model, the object was launched from a height of 0 feet and reached its maximum height of 1,600 feet 10 seconds after it was launched. Based on the model, what was the height, in feet, of the object 7 seconds after it was launched?

139. [December 2024]

$$(x - k)^2 = (k - 4a)(x - k)$$

In the given equation, a and k are constants, where $k > 4a$. The sum of the solutions to the equation is $3k + 35$. What is the value of a ?

140. [March 2025]

$$x + 7 = 13$$

$$y = 5x^2 + 5$$

At what point (x, y) do the graphs of the equations in the given systems intersect?

141. [March 2025]

$$x(x + 4) - 140 = 3x(x - 10)$$

What is the sum of the solutions to the given equation?

- A) 6
B) 10

- C) 14
D) 17

142. [March 2025]

$$-3x^2 - 7x + 1 = 0$$

What is the greatest solution to the given equation?

- A) $-\frac{7}{3} + \frac{\sqrt{61}}{3}$
B) $-\frac{7}{3} - \frac{\sqrt{61}}{3}$
C) $-\frac{7}{6} + \frac{\sqrt{61}}{6}$
D) $-\frac{7}{6} - \frac{\sqrt{61}}{6}$

143. [March 2025]

An object is launched from a height of 121 feet above the ground. A quadratic function f models the height above the ground, in feet, of the object t seconds after it is launched. According to the model, 2 seconds after the object is launched, it reaches a maximum height of 185 feet above the ground. Which equation defines f ?

- A) $f(t) = -16(t + 185)^2 - 2$
- B) $f(t) = -16(t - 185)^2 + 2$
- C) $f(t) = -16(t + 2)^2 - 185$
- D) $f(t) = -16(t - 2)^2 + 185$

144. [March US 2025]

$$x(13x + 27) + c = 5$$

In the given equation, c is a constant. A solution to the equation is $-\frac{27 + \sqrt{261}}{26}$. What is the value of c ?

- A) 14
- B) 9
- C) 5
- D) 4

145. [March US 2025]

The function h is defined by $h(x) = -\sqrt{x^2 + bx + c}$, where b and c are constants. In the xy -plane, the graph of $y = h(x)$ contains the points $(2, 0)$ and $(0, -\sqrt{358})$. If $h(m) = 0$, what is the greatest possible value of m ?

146. [March 2025]

The function f is a quadratic function. In the xy -plane, the graph of $y = f(x)$ has a vertex at $(1, 7)$ and passes through the points $(2, 53)$ and $(-1, 191)$. What is the value of $f(-2) - f(0)$?

- A) 145
- B) 237
- C) 368
- D) 421

147. [March 2025]

$$y = -0.5$$

$$y = x^2 + 8x + a$$

In the given system of equations, a is a positive integer constant. The system has no real solutions. What is the least possible value of a ?

- A) 13
- B) 14
- C) 15
- D) 16

148. [March 2025]

$$y = x - c$$

$$y = -6(x - 12)^2$$

In the given system of equations, c is a constant. The system has two distinct real solutions. Which of the following could be the value of c ?

- A) 7
- B) 11
- C) $\frac{287}{24}$
- D) 17

149. [May 2025]

$$f(x) = x^2 - 4x - 320$$

The function f is defined by the given equation. Which of the following equivalent forms of the equation displays the minimum value of the function as a constant or coefficient?

- A) $f(x) = x^2 - 4(x + 80)$
- B) $f(x) = (x - 2)^2 + (-324)$
- C) $f(x) = x(x - 4) + (-320)$
- D) $f(x) = (x + 16)(x - 20)$

150. [May 2025]

$$3x^2 - 5x - 5 = 0$$

What is the greatest solution to the given equation?

- A) $\frac{5}{6} - \frac{\sqrt{85}}{6}$
- B) $\frac{5}{6} + \frac{\sqrt{85}}{6}$
- C) $\frac{5}{3} - \frac{\sqrt{85}}{3}$
- D) $\frac{5}{3} + \frac{\sqrt{85}}{3}$

151. [May 2025]

$$y = x^2 + 17x + 4$$

$$y = x + 4$$

The graphs of the equations in the given system intersect at the point (x, y) in the xy -plane. What is a possible value of x ?

- A) -16
- B) -15
- C) 13
- D) 15

152. [May 2025]

Which quadratic equation has exactly one distinct real solution?

- A) $x^2 - 16 = 0$
- B) $x^2 + 16 = 0$
- C) $x^2 - 16x + 56 = 0$
- D) $x^2 - 16x + 64 = 0$

153. [May 2025]

$$f(x) = -0.0038400x^2 + 15.236x - 15,103$$

The model shown gives the predicted average Arctic sea ice area $f(x)$, in millions of square kilometers, for September of each year x , where x represents a year and $1979 \leq x \leq 2012$. Based on the model, what is the positive difference, in millions of square kilometers, between the predicted average Arctic sea ice area for September of the year 1991 and the predicted average Arctic sea ice area for September of the year 1992? (Round your answer to the nearest thousandth.)

154. [May 2025]

$$\frac{1}{cx} = \frac{x}{96} + \frac{1}{c}$$

In the given equation, c is a constant. If the equation has exactly one solution, what is the value of c ?

155. [May 2025]

$$\begin{aligned} y &= 0 \\ y &= 8(x^2 - 36) \end{aligned}$$

Which ordered pair (x, y) is a solution to the given system of equations?

- A) $(0, 6)$
- B) $(0, 36)$
- C) $(6, 0)$
- D) $(36, 0)$

157. [May 2025]

$$h(t) = -16t^2 + b$$

The function h estimates an object's height, in feet, above the ground t seconds after the object is dropped, where b is a constant. The function estimates that the object is 40.96 feet above the ground when it is dropped at $t = 0$. How many seconds after being dropped does the function estimate the object will hit the ground?

158. [May 2025]

$$x^2 - 28x = 0$$

Which of the following is a solution to the given equation?

- A) 56
- B) 28
- C) 14
- D) $\sqrt{28}$

156. [May 2025]

$$(x - 42)^2 = 1$$

What is the sum of the solutions to the given equation?

- A) 42
- B) 43
- C) 84
- D) 85

159. [May 2025]

$$x^2 - 15x + 12 = 0$$

What is the sum of the solutions to the given equation?

- A) 15
- B) 12
- C) 3
- D) 0

160. [May 2025]

The function $f(x) = \frac{1}{7}(x - 6)^2 + 4$ gives a toy car's height above the ground $f(x)$, in inches, x seconds after it started moving on an elevated track, where $0 \leq x \leq 10$. Which of the following is the best interpretation of the vertex of the graph $y = f(x)$ in the xy -plane?

- A) The toy car's minimum height was 4 inches above the ground.
- B) The toy car's minimum height was 6 inches above the ground.
- C) The toy car's height was 4 inches above the ground when it started moving.
- D) The toy car's height was 6 inches above the ground when it started moving.

161. [May 2025]

The function h is defined by

$$h(x) = -\sqrt{x^2 + bx + c},$$

where b and c are constants. In the xy -plane, the graph of $y = h(x)$ contains the points $(2, 0)$ and $(0, -\sqrt{334})$. If $h(m) = 0$, what is the greatest possible value of m ?

162. [June US 2025]

$$-x^2 + bx - 64 = 0$$

In the given equation, b is an integer. The equation has no real solutions. What is the least possible value of b ?

163. [June 2025]

$$x^2 - 2 = 23$$

Which value is a solution to the given equation?

- A) -23
- B) -21
- C) -2
- D) 5

164. [June 2025]

$$m(t) = -0.0274 \left(\frac{t}{7} \right)^2 + 7.3873 \left(\frac{t}{7} \right) + 75.032$$

The function m gives the predicted body mass $m(t)$, in kilograms (kg), of a male giraffe t days after it was born in a wildlife reserve, where $t \leq 390$. Which of the following is the best interpretation of the statement " $m(140)$ is approximately equal to 212" in this context?

- A) The predicted body mass of a male giraffe was approximately 140 kg 212 days after it was born.
- B) The predicted body mass of a male giraffe was approximately 212 kg 140 days after it was born.
- C) The predicted body mass of a male giraffe was approximately 212 kg $\frac{140}{7}$ days after it was born.
- D) The predicted body mass of a male giraffe was approximately $\frac{140}{7}$ kg 212 days after it was born.

165. [June 2025]

$$(2x^2 + 3x - 3) - (7x^2 + 2x)$$

The given expression is equivalent to

$$-5x^2 + ax - 3,$$

where a is a constant. What is the value of a ?

166. [June 2025]

$$y = -3x^2 - 40$$

$$y = qx - 37$$

In the given system of equations, q is a positive constant. The graphs of the equations in the given system intersect at exactly one point, (x, y) , in the xy -plane. What is the value of x ?

A) -77

B) -43

C) -6

D) -1

168. [June 2025]

$$f(x) = (x - a)(x - b)$$

The function f is defined by the given equation, where a and b are integer constants. If $f(32) > 0$, $f(35) < 0$, and $f(38) > 0$, which of the following could be the value of $a + b$?

A) 35

B) 37

C) 67

D) 69

169. [June 2025]

$$\sqrt{k - x} = 42 - x$$

In the given equation, k is a constant. The equation has exactly one real solution. What is the minimum possible value of $4k$?

167. [June 2025]

$$x^2 - 11x - 3 = 0$$

What is a solution to the given equation?

A) $\frac{11 + \sqrt{(-11)^2 - 4(1)(-3)}}{2(-3)}$

B) $\frac{11 + \sqrt{(-11)^2 - 4(1)(-3)}}{2(1)}$

C) $\frac{3 + \sqrt{(-3)^2 - 4(1)(-11)}}{2(-3)}$

D) $\frac{3 + \sqrt{(-3)^2 - 4(1)(-11)}}{2(1)}$

170. [June 2025]

$$z(kz - 112) = 0$$

In the given equation, k is an integer constant. If the equation has two distinct real solutions, what is the greatest possible value of k ?

171. [June US 2025]

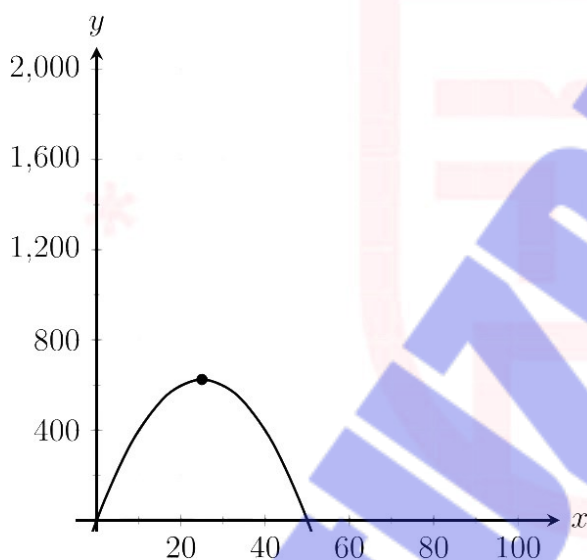
$$x^2 + y = 21$$

$$-x + y = 1$$

The solutions to the given system of equations are of the form (x, y) . What is a possible value of x ?

- A) -5
- B) -4
- C) -1
- D) 21

172. [June US 2025]



The graph gives the estimated revenue, in dollars, for a company when the price of the company's product is x dollars. What does the y -value of the vertex of the graph represent in this context?

- A) The price, in dollars, of the company's product
- B) The minimum estimated revenue, in dollars
- C) The maximum estimated revenue, in dollars
- D) The estimated revenue, in dollars, when the price of the company's product is 0 dollars

173. [June US 2025]

$$w^2 + 8w - 28 = 0$$

Which of the following is a solution to the given equation?

- A) $4 - 2\sqrt{11}$
- B) $2\sqrt{11}$
- C) $\sqrt{11}$
- D) $-4 + 2\sqrt{11}$

174. [June US 2025]

The function $f(x) = \frac{1}{9}(x - 6)^2 + 3$ gives a toy car's height above the ground $f(x)$, in inches, x seconds after it started moving on an elevated track, where $0 \leq x \leq 10$. Which of the following is the best interpretation of the vertex of the graph of $y = f(x)$ in the xy -plane?

- A) The toy car's minimum height was 3 inches above the ground.
- B) The toy car's minimum height was 6 inches above the ground.
- C) The toy car's height was 3 inches above the ground when it started moving.
- D) The toy car's height was 6 inches above the ground when it started moving.

175. [June US 2025]

Function f is a quadratic function. The graph of $y = f(x)$ in the xy -plane has a vertex at $(4, -6)$, contains the point $(3, 8)$, and has a y -intercept at $(0, a)$. The graph of $y = 5 + f(x)$ has a y -intercept at $(0, b)$. What is the positive difference between a and b ?

178. [June US 2025]

$$18x^2 - 24x + c = 0$$

In the given equation, c is a constant. The equation has exactly one solution. What is the value of c ?

176. [June US 2025]

At what value of x does the graph of

$$y = x^2 + 18x - 23$$

reach its minimum in the xy -plane?

A) -23

B) -9

C) 9

D) 18

$$\frac{x+1}{5x^2} = \frac{k}{x}$$

In the given equation, k is a constant. The solution to the given equation is $\frac{1}{224}$. What is the value of k ?

179. [June US 2025]

180. [June US 2025]

177. [June US 2025]

$$x + 5 = 11$$

$$y = 3x^2 + 3$$

At what point (x, y) do the graphs of the equations in the given system intersect?

A) $(6, 108)$

B) $(6, 111)$

C) $(11, 3)$

D) $(11, 366)$

$$y = x - c$$

$$y = -4(x - 6)^2$$

In the given system of equations, c is a constant. The system has two distinct real solutions. Which of the following could be the value of c ?

A) 1

B) 5

C) $\frac{95}{16}$

D) 11

181. [November 2023]

The function $f(x) = \frac{1}{7}(x - 5)^2 + 6$ gives the toy car's height above the ground $f(x)$, in inches, x seconds after it started moving on an elevated track, where $0 < x \leq 10$.

Which of the following is the best interpretation of the vertex of the graph of $y = f(x)$ in the xy -plane?

- A) The toy car's minimum height was 6 inches above the ground.
- B) The toy car's minimum height was 5 inches above the ground
- C) The toy car's height was 6 inches above the ground when it started moving
- D) The toy car's height was 5 inches above the ground when it started moving

182. [November 2023]

x	y
20	-20
24	12
28	-20

The table shows three values of x and their corresponding values of y , where $y = f(x) + 6$ and f is a quadratic function. What is the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane?

183. [May 2024]

The product of a positive number x and the number that is 7 less than x is equal to 228. What is the value of x ?

184. [June 2023]

The graph of the quadratic function $y = f(x)$ in the xy -plane intersects the x -axis when $x = 39$ and when $x = p$, where p is a constant. The maximum value of $y = f(x)$ occurs at the point $(14, m)$, where m is a constant. What is the value of p ?

185. [November 2023]

$$y = -0.5$$

$$y = x^2 + 10x + a$$

In the given system of equations, a is a positive integer constant. The system has no real solutions. What is the least possible value of a ?

186. [May 2025]

A club plans to sell tote bags. The club members estimate they will sell 80 tote bags when the bags are priced at \$9 each. For every price increase of \$1, they estimate they will sell 8 fewer bags. What is the estimated revenue, in dollars, when the bags are priced at \$12 each? (revenue = price $\times$ number of bags sold)

187. [June 2024]

The product of a positive number x and the number that is 1 less than x is equal to 342. What is the value of x ?

- A) 0.5
- B) 19
- C) 341
- D) 343

188.

For the parabola $px^2 + qx + r$, the x -intercepts are $(10, 0)$ and $(m, 0)$.

If $f(25) = f(3)$, what is the value of m ?

- A) 16
- B) 18
- C) 20
- D) 22

189.

The function $g(z) = pz^2 + qz + r$ has zeros at $z = 1$ and $z = 7$.

Suppose p is an integer greater than 1. Which of the following could be the value of $p + q$?

- A) -14
- B) -16
- C) -8
- D) -24

190.

When the quadratic function h is graphed in the xy -plane, where $y = h(x)$, its vertex is $(2, -3)$. One of the x -intercepts of this graph is $(\frac{11}{4}, 0)$. What is the other x -intercept?

- A) $(\frac{3}{4}, 0)$
- B) $(-1, 0)$
- C) $(\frac{5}{4}, 0)$
- D) $(-\frac{5}{4}, 0)$

191.

A quadratic function models the height, in meters, of an object launched from an elevated surface. Measurements show that at $t = 8$ s, the object is 254 m high, and at $t = 12$ s, it is 270 m high.

If the object's height at launch ($t = 0$) was 30 m, what is its height at $t = 17$ s?

- A) 140
- B) 180
- C) 200
- D) 220

192.

A quadratic function models the height, in meters, of an object above the ground as a function of time t , in seconds, after launch.

The object is launched from the ground (0 m) and reaches its maximum height of 400 m at $t = 10$ s.

Based on the model, what is the height of the object at $t = 14$ s?

- A) 256
- B) 288
- C) 336
- D) 384

193.

A quadratic function h has vertex $(2, -3)$ and passes through the points $(0, 5)$ and $(3, -1)$.

What is $h(4) - h(1)$?

- (A) -10
- (B) -8
- (C) 4
- (D) 6

194.

A quadratic function p has vertex $(3, -4)$ and passes through $(2, -6)$. If

$$p(x) = -2(x - 3)^2 - 4,$$

then $a = p(0)$. If

$$T(x) = p(x) - p(0),$$

then what is the positive difference between a and $T(0)$?

195.

$$px^2 + 60x + d$$

has at least one real root and is divisible by the linear factor $rx + s$ (with p, r, s, d all positive integers).

What is the greatest possible value of pd ?

- A) 820
- B) 864
- C) 882
- D) 900

196.

$$4x^2 + 22x + d,$$

where **4**, **22**, and **d** are positive constants. Suppose $x + 3$ is a factor of this expression **and** the quadratic has real solutions.

What is the greatest possible integer value of d ?

- A) 28
- B) 29
- C) 30
- D) 31

197.

Which of the following expressions is divisible by $x + 2$?

- (A) $4x^2 + 20x + 24$
- (B) $4x^2 + 28x + 24$
- (C) $4x^2 + 32x + 24$
- (D) $4x^2 + 36x + 24$

198.

The function f is defined by the equation

$$f(x) = (x - a)(x - b),$$

where a and b are integer constants. If $f(2) > 0$, $f(5) < 0$, and $f(9) > 0$, what is **one possible value** of $a + b$?

199.

The function g is defined by the equation

$$g(x) = (x - a)(x - b),$$

where a and b are integer constants. If $g(1) > 0$, $g(4) < 0$, and $g(6) > 0$, what is **one possible value** of $a + b$?

200.

The function f is defined by $f(x) = ax^2 + bx + c$, where a , b , and c are constants. The graph of $y = f(x)$ in the xy -plane passes through the points $(10, 0)$ and $(-3, 0)$. If a is an integer greater than 1, which of the following could be the value of $a + b$?

- (A) -12
- (B) -6
- (C) 7
- (D) 8

201.

In the xy -plane, a parabola has vertex $(9, -14)$ and intersects the x -axis at two points. If the equation of the parabola is written in the form $y = ax^2 + bx + c$, where a , b , and c are constants, which of the following could be the value of $a + b + c$?

- A. -23
- B. -19
- C. -14
- D. -12

202.

The equation of a parabola is written in the form $y = ax^2 + bx + c$, where a , b , and c are constants. When graphed in the xy -plane, the parabola has vertex $(-1, 4)$ and does not intersect the x -axis. Which of the following could be the value of $a - b - c$?

- (A) -5
- (B) -4
- (C) 0
- (D) 4

203.

If the vertex of a parabola is at the point $(1, 24)$ and the quadratic can be written in the form $-a(x^2) + bx + c$ and a , b , and c are all positive integers, what is the maximum value of b ?

204.

In the xy -plane, a parabola has a vertex $(7, 12)$ and intersects the x -axis at two points. If the equation of the parabola is written in the form $y = ax^2 + bx + c$, where a , b , and c are constants, which of the following could be the value of $a + b + c$?

- A) 10
- B) 14
- C) 18
- D) 36

205.

In the xy -plane, a parabola has a vertex $(5,1)$ and intersects the x -axis at two points. If the equation of the parabola is written in the form $y = -ax^2 + bx + c$, where a , b , and c are constants and $a > 0$, which of the following could be a value of c ?

- A) -5
- B) 2
- C) 3
- D) 9

206.

Quadratic function h is defined by $h(x) = ax^2 - 2x + c$, where a and c are constants. The graph of $y = h(x)$ in the xy -plane is a parabola that opens downward and has a vertex of (h, k) , where $k > 0$. If $h(1) = h(-13)$, which of the following must be true?

- I. $-1 < a < 0$
- II. $c > 0$

- ☐ A I only
- ☐ B II only
- ☐ C I and II
- ☐ D Neither I nor II

207.

$$f(x) = ax^2 + 4x + c$$

In the given quadratic function, a and c are constants. The graph of $y = f(x)$ in the xy -plane is a parabola that opens upward and has a vertex at the point (h, k) , where h and k are constants. If $k < 0$ and $f(-9) = f(3)$, which of the following must be true?

- I. $c < 0$
- II. $a \geq 1$

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

208.

$$k(x) = rx^2 + 8x + t$$

In the given quadratic function, r and t are constants. The graph of $y = k(x)$ in the xy -plane is a parabola that opens downward and has a vertex at the point (u, v) , where u and v are constants. If $v > 0$ and $k(-1) = k(3) > 0$, which of the following must be true?

- I. $t > 0$
- II. $r < -8$

- ☐ A I only
- ☐ B II only
- ☐ C I and II
- ☐ D Neither I nor II